

OnLine Test System Applied in Routing Protocol Test

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Abstract

Based on the analysis and the perception of routing protocols, this paper presents an online test method applicable for the routing protocol test. This method tests the protocol implementation through online observation. It can be performed in production field without interfering to the network and possesses quite a lot other beneficial features. With the key techniques were worked out, the target and the framework of an online test system (OLTS) are put forward. The OLTS exploits the state synchronization algorithm to test the protocol state machine, tests the exchange and the manipulation of the routing information through the topology analysis and the internal process simulation. OLTS provides an intuitionistic and friendly graphical user interface and good flexibility. Now the prototype of the OLTS has been implemented and the test activity about BGP, OSPF and RIP has been performed.

1. Introduction

Protocols are the soul of the network. Correct and efficient implementations of protocols are the crucial factor of the network. With the development of the Internet, the scale of the network is booming up exponentially with a much more complicated architecture. The routing protocols are playing an important role for the network performance. The protocol test is an effective means to guarantee the quality of the protocols implementations and the test of routing protocols is becoming both a hot spot

for researchers and a necessary phrase for network equipment manufactures.

Now the most commonly used routing protocols include RIP [1], OSPF [2] and BGP [3]. In our study on the test of Internet routing protocols, we analyzed the features of routing protocols and the limitation of current test system and test method. With more considerations of the requirements from the practical application, we adopted an on-line test method to test routing protocols. Based on this, the framework of an OnLine Test System (OLTS) was work out. The key theories and techniques needed to develop the system were studied. The state synchronization algorithm [4] is exploited to test the protocol state machine. Through the topology analysis [5] and the internal process simulation, the routing information exchange and manipulation are tested. The extensibility and the universality are taken care of from the design stage. During the development of the OLTS, the OOP technique is exploited to implement the Virtual Network Interface (VNI). TCL/Tk is adopted to draw the GUI. The prototypes of the OLTS are implemented on both a Sun Ultra 1 Solaris platform and a x86-Linux platform separately to promote the usability.

This paper is composed of seven sections. Section 2 discusses the features and the test requirements of routing protocols. Section 3 briefs the online test method and lists the key techniques to be solved. Section 4 explains the goal of the design and the system framework. Section 5 explains the features of the OLTS implementation. Section 6 describes the practice of the OLTS. Finally, related work and conclusions are given.

2. Features and Test Requirements of Routing Protocols

2.1. Features of Routing Protocols

The progresses achieved so far in the protocol test field mainly focus on the test of communication protocols, such as TCP/IP. While routing protocols possess different features. Firstly, the goal of a routing protocol is to supply the information that is needed to do routing [1], which is a basic summary of its function. The concerns of Routing protocols are the information about the network topology without the requirement of the "best effort" transmitting. Every routing protocol has the consideration of the reduction of both the interaction frequency and the amount of data, especially avoiding bursty data or heavy network bandwidth occupation when updating the routing information. All the routing protocols have to take advantage of the service offered by communication protocols, for example, BGP exploits TCP. The difference of the service provider has some effect on the feature of the above protocol. Determining corresponding communication interface is the precondition of the test.

An outstanding character of routing protocols is their properties of distributed systems. There are diversities in the complexity among BGP, RIP and OSPF while they all possess the common features of distributed systems. All interior and inter Autonomous System routers make up the backbone of the Internet and the network frameworks of all levels. Even to each router, there may be a separate BGP-4 process (with its own state-machine and database) running on each port.

For routing protocols, there are several important databases both in both the concept and the implementation where the routing information is stored. Routing table is the most common database, directly driving the routing. Other databases are used in the processing of the routing information. For example, in BGP-4, the RIB (Routing Information Base) is composed of three parts: Adj-RIB-In, Loc-RIB and Adj-RIBs-Out, recording the received,

processed routing information and that to be sent out.

There are quite a lot of internal processes in routing protocols. They are much complicated than those of communication protocols, such as the sliding window protocol in TCP. These processes dominate the correctness and the performance of routing protocols. For example, the internal processes of BGP-4 are called Decision Processes, which is composed of three stages. In first stage, the preference of one route is calculated. Then, route selection is done in the second stage. The third stage is the Route Dissemination. These three stages may be activated in turn and also may be mutex due to database operations. In each stage of the process, large amount of rules and policies are applied, including the selection of the most favorite route, breaking ties, dealing with the overlapping routes, etc.

Besides, there are still differences between the level compared with communication protocols. The N level communication protocol takes advantage of the service offered by the N -1 level and offers service for the N + 1 level. There are no "upper" protocols to routing protocols. But the interaction of routing protocols is much more complicated. There are two main interactions. Not only do BGP Peers talk to each other; BGP also interacts with other routing protocols like OSPF locally (in the same router).

As to communication protocols, the state machine is the basis from the test suite design till the coverage analysis. Since the communication protocol always doesn't concern the content of the user payload of upper layer. Its main task is just to transmit this data. It cares more about the link establishment and maintenance. Therefore, it is reasonable to take the state machine as the basis. These are the descriptions of state machines of routing protocols (of course, the state machine of RIP is so simple that it only contains one state), but they only describe a little part of their activities. Most of the features are the activities after the routing protocol enters a "stable" state (as for BGP, it is the state called Established).

2.2. Test Requirements

The test of routing protocols should have all above features considered. The scope should include (1) the structure of the protocol packet data unit (PDU), (2) the protocol state machine, (3) the interoperability, (4) the exchange and processing of the routing information and (5) the performance evaluation.

The test of the PDU structure will check the basic function of composing the packets. This correctness directly affects the conformance and the interoperability of protocol implementations. Not only should the packets sent out by IUT (implementation under test) be checked, its ability to deal with incorrect PDU should also be tested. The state machine is the description of protocol states transitions, which is an important proportion of the protocol. The transitions and paths (including the processing of untimely PDU) should be tested. The interoperability test will ensure different routers can interact correctly.

The interaction and the processing of the routing information are import scopes to be covered in the test. When the topology of the network changes, the routing protocols should discover and propagandize such conditions. Corresponding test should include the processing and transmitting of this information, the maintenance and updating of databases, and the correctness and efficiency of the internal processes.

The criterion of the routing protocol performance is far different from that of the communication protocols which is always the speed of data transmitting and that of routers which is the ratio of packet forwarding speed. Quick response to changes of the network topology and efficient data exchange are important performance indexes of routing protocols.

3. Online Test Method and Key Techniques

3.1. Brief of Online Test Method

Current test methods in common use [6] all perform the test activity based on the test suite. The test system injects

the input to IUT, accepts its response (output) and gives a verdict. All these methods are called active test methods. These methods are always used in a test environment. They can't be used in an actual network environment. Therefore they are also "off line" methods. With these methods, the coverage and the duration of the test are determined by the test suite. These methods always require special APIs, which restricts the system extensibility and the flexibility. Since the test system will manage the organization and the execution of all the test cases, large amounts of system resource is occupied, leading to the low efficiency and the inability for testing in a high-speed network.

While the online test method is a passive and online test method. The model is shown in figure 1. Where:

IUT: Implementation Under Test. IUT₁ and IUT₂ communicate with each other.

DC: Data Channel.

OLT: Online Tester. It monitors sessions between IUT₁ and IUT₂ without sending out anything, gives verdicts and provides the information for the fault management periodically or on demand.

Compared with the active testing, one of the remarkable features of the online test is its simple structure, while possessing a more feasible and powerful test ability to perform the conformance test, the interoperability test and even the analysis on performance. Besides, it allows more IUTs and instances of OLTs, which makes it able to work in a distributed way.

As to the network management, since it works online silently, the online test can be performed in the production field without interfering with the network. And it can almost work there forever, taking care of the network at any time and dealing with abnormalities that really happen. It is independent of the SNMP so that it can go ahead with irredeemably corrupted routers.

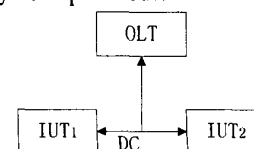


Figure 1. Model of the online test method

3.2. Key Techniques

3.2.1. Test Protocol State Machine Online. The protocol state machine is an important aspect of the protocol test. The online test must also be able to test it. Therefore, the state determination must be solved first, that is, determining the states of the peers from the session observed. Then the correctness of the state machine implementation can be checked. Since the online test only observes passively, it is quite different from the traditional "injecting input/stimulation and observing output/response" way. The state synchronization algorithm [4] is right to identify the peer states through the online observation. Necessary definitions will be given first to describe the activity of the online observation.

Def. 1. Session Interaction. A Session Interaction ("SI" for short) is a 4-tuple

$$SI = \{t, PDU, \text{Sender}, \text{Receiver}\}$$

Where t is the time-point when we observe that the PDU is sent from **Sender** to **Receiver**. To the **Sender**, it generates an event of sending out a packet, to the **Receiver**, it encounters an event of receiving a packet.

Def.2. Session Interaction Sequence. Several successive SIs make up a Session Interaction Sequence (SIS for short), that is:

$$SIS = \{SI_1, SI_2, \dots, SI_i, SI_{i+1}, \dots, SI_j\}$$

$$\text{Where } SI_m = \{t_m, PDU_m, \text{Sender}_m, \text{Receiver}_m\}$$

$$m \in [1 \dots j], \quad t_i < t_{i+1}, \quad j - 1 \geq i \geq 1$$

$$\forall SI_i, SI_{i+1} \quad t_i < t_{i+1} \quad \neg(\exists SI_k \text{ satisfying } t_i < t_k < t_{i+1})$$

Also, it means a sequence of events of the **Sender** and **Receiver**.

Then comes the definition of the State Synchronization.

Def. 3. State Synchronization. A State Synchronization is a transform T

$$T : SIS \rightarrow (t_j, \text{State}), \text{State} = (\text{State}_s, \text{State}_r),$$

$$\text{State}_s, \text{State}_r \in \text{FSM}$$

Where **State** is a vector composed of the states of the peers $\text{State}_s, \text{State}_r$. And the FSM is the Finite State Machine of the protocol.

Detailed description and analysis of this algorithm can be referred to [4].

3.2.2. Test of Routing Information Exchange and Manipulation. Test of routing information exchange and manipulation is another key technique to develop the OLTS. One of the methods exploited is the topology analysis. The main idea is that correct routing information must describe the network topology correctly. By observing the session online, the routing information is collected, parsed and analyzed. Then the topology is constructed based on this information and compared with the actual topology. Thus the abnormality in the routing information exchange and manipulation can be detected. The detailed description and discuss of this method and its application can be referred to [5].

Although we can employ some common method such as the topology analysis in the routing protocol test, generally speaking, there is no common way to test the internal processes of routing protocols. We can only use specific method for specific test purpose. For example, BGP-4 routing information operation consists of route withdrawing and route advertisement. A route is presented in the form of a 2-tuple $\langle \text{Length}/\text{IP-Prefix} \rangle$. It is always a network whose address is specified by the left Length bits. For a route A advertised previously, the relation between the later withdrawal or the advertisement of a route B can be cases **a**, **b**, **c** and **d** in figure 2, but can't be case **e** (the proof is simple and omitted here). To cases **b**, **c** and **d**, analysis will be performed to test the routing information manipulation of BGP-4. Take case **c** as an example and suppose B is a withdrawn route. Since network B is a subnet of network A and the route to network B is invalid (withdrawn route), while the route to A\B (other part in A excluding B) may remain unchanged. If we test these cases, corresponding process and analysis are needed.

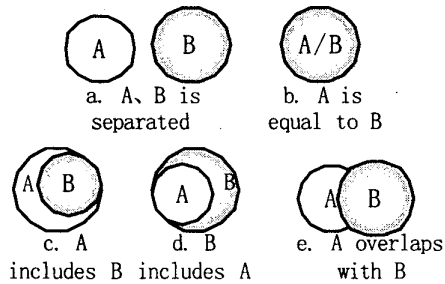


Figure 2. Relations between two routes

4. Design Target and Framework of OLTS

4.1 Design Target

The test system is a realization of the test method, making the test method function. The OnLine Test System should possess such features and abilities below:

- (1) Capable to perform the conformance test, the interoperability test and the performance test.
- (2) Able to test distributed protocols. Currently, distributed systems and distributed protocols are widely employed. This is also an import field of the protocol test.
- (3) Able to test high-speed network protocols. This is also the demand from the network development.
- (4) With an appropriate universality. Overly emphasizing the universality will always lead to a slow trite system; On the other hand, neglecting the universality will not only take large quantity of the manpower and money to do repeated work, which makes the test always fall behind the technique development. It will also have negative effect on the test standardization and the application. Therefore, determining a proper universality is also an important task for the system design.
- (5) Able to test in production field. This will be the most appealing factor of the OLTS.
- (6) Able to cooperate with current test system. OLTS has its own advantage, but it also has limitations of its own. It can only monitor the sessions but can't send out specific packet as the input. Therefore, what the session involved determines all the testable scope. The frequently used or seldom used sessions can be tested pertinently. It is also desirable for OLTS to generate some test suite based on

the sessions observed and provide it to current test system.

In order to make full functions out of the OLTS and achieve the design target, it is very important to refer to the successful experience of current test systems and set aside enough space for them to cooperate with each other. Standardization is another factor needs careful consideration for evaluation. Still, modularization should be taken care of for the implementation and the extensibility.

4.2. System Framework

The framework of OLTS is shown in figure 3. There are five main components. The input component includes the protocol state machine description module, the protocol packet structure description module and the set condition module. The communication component includes the communication interface module and the condition check and filter module. The communication interface module shields the difference among communication protocols of different layers. The condition check and filter module further processes the data collected and provides the test component with a uniform interface. The test component is the kernel of the OLTS, including the protocol state machine test module, the internal process simulation module and the statistics module. The protocol state machine test module also exploits the state synchronization algorithm to test the state machine. The statistics module counts and analyzes the frequency and the quantity of data exchange, judges its correctness and gives performance indexes. The internal process simulation module behaves just like the internal process of the protocol. The cooperation component is used to organize multiple OLTSs instances to perform the test. The test result module, the LOG module and other modules belong to the conclusion and presentation component, in charge of composing a test report, providing a graphic presentation and generating test suites.

For different protocol under test, specific communication interface module is coded, descriptions of the protocol state machine and the protocol packet

structure are input. Necessary internal process simulation module is applied or implemented.

5. Implementation of OLTS

OLTS is implemented on a Sun Ultra 1 - Solaris and a x86-Linux separately in order to widen its application and verify its portability. With an OOP of C++, OLTS is modularized, reusable and portable. TCL/Tk is used to support the GUI.

Parts related to specific protocols include the protocol state machine description and the protocol packet structure description. The protocol state machine is described with formal description language. To packet structure, the Abstract Syntax Notation (ASN.1) is exploited. The descriptions are all written in plain text, easy to read and edit.

As to the communication interface, specific codes will be written for different communication media to shield the difference of the lower level. For example, the interface has been implemented for the most popular Ethernet, which is able to capture all the data of the upper level protocols. On x86-Linux, Packet socket is exploited and DLPI (Data Link Provider Interface) operating /dev/le on Sun Ultra 1 - Solaris. For other communication media, there is no difficulty encountered so far (even for a non-broadcast point to point link). Besides, DLPI supports operations on X.25, LAPB, SDLC, ISDN, Ethernet, FDDI,

Token ring, Token Bus, etc. Transfer to higher speed network equipment will not affect the code of the OLTS. The communication interface has two main processes. The main process forks a child process that extracts protocol frames of interest from the communication channel, puts them in the queue and maintains them. The main process takes out the packets in turn and works on them. By this means, the possibility of packet loss can be avoided to the maximum extent and the main process is released to do other much more important work. On the other hand, the correlation of specific protocol is further reduced, which increases the universality of OLTS and its applicability in a high-speed network. Based on this, the virtual network interface (VNI) is further implemented. VNI aims to provide a uniform interface that the main process can operate with. When new communication media are introduced, what needed is just mapping its interface to the uniform interface. No modification to other parts of the system is needed. For multiple communication interfaces, an interface selection function is implemented. The internal process simulation is a module stack; the modules can be pushed into the stack for different protocols and test purposes.

OLTS also provides a GUI to promote usability. Through the GUI, the user can set the conditions, parameters and observe the results. The GUI includes a real time display and a playback. The former can display the test procedure and results in real time and the latter is

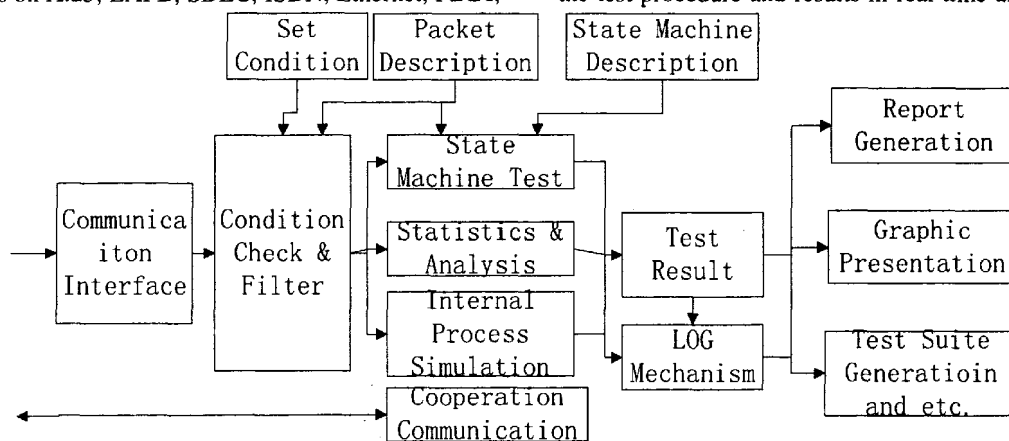


Figure 3. Framework of the OLTS

convenient for detailed analysis and the fault location.

For consideration of testing in distributed system, the Simple Cooperation Protocol (SCP) [7] is defined for the cooperation component, which enables multiple instances of OLTS to work together in different places of the network. The SCP exploits the reliable transmission of TCP and the protocol function is very simple to guarantee the reliability and the efficiency. SCP defines three main operations: **Command**, **Result** and **Trap**. One instance of the OLTS requires another to accomplish some specified task (now it is to collect the information needed) through **Command**. The latter feeds back with **Result**. **Trap** is used to advertise some specific information (e.g. exceptions).

6. Practice of OLTS

OLTS has worked on the backbone of the CERNET [8]. Since the OLTS only listens silently online, it has no inference to the network. The BGP-4 has been tested, including the protocol packet structure, the protocol state machine, the exchange and processing of the routing information. Preliminary analysis and evaluation have been made about the backbone performance. Detailed report will be released later. A graphical presentation generated by OLTS is shown in figure 4. This figure describes the topology based on the routing information collected. Compared with actual network topology, this generated one can be used to check the correctness of the routing information manipulation.

Besides, we have tested some medium-end and high-end commercial routers. Beneficial results have been achieved. Some errors of the implementations are found. One case is shown in figure 5, which records one session between a commercial router and a prototype Router II.

166.111.69.91 is one port address of the commercial router and 166.111.69.75 is that of the Router II. The session shows that due to the incorrect implementation of BGP-4 of Router II, UPDATE packets are sent out repeatedly instead of OPEN after the TCP connection is established. But the response of the commercial router is

not correct either. After it responses with a NOTIFICATION packet (line 2 in figure 5), it answers with the same UPDATE (line 5 in figure 5), which is contradict with the protocol specification and leads to problems of the conformance and the interoperability and extra cost of the network bandwidth.

7. Related Work and Conclusions

Protocol Integrated Test system [9] is the second generation of test system we developed and has performed large quantity of test on communication protocols. It is an off line active test system. OLTS has referred to the successful experience of PITS a lot. Currently, there are some commercial test systems, such as the HP Router Tester [10], the NetCom Systems Smartbits [11], etc. They mainly focus on the performance, such as the throughput, delay, jitter, etc. Such test system sends out large quantity of data to IUT and measures the parameters about the performance.

Despite of the importance of routing protocols, the test activity is still very limited. Reference [12] presents the test of IP routing protocols with the test environment generated by the software tool SOCRATES. Reference [4] is the intelligent online BGP-4 analyzer, which provides the design of OLTS with valuable references.

In the study of routing protocol test, the online test method is adopted aimed at the features of routing protocols to overcome the deficiency of the current test system. This test method is simple in architecture and flexible, able to test in production field without interference to the network. With the key techniques worked out, the framework of the OLTS was designed. OLTS is implemented on Sun Ultra 1-Solaris and x86-Linux separately. Through the test activity, and the work on the backbone of the CERNET, valuable results have been gotten. The correctness and the validity of this test method, the key techniques and the OnLine Test System itself are proved. In the future, our study and testing of routing protocols will be going on.

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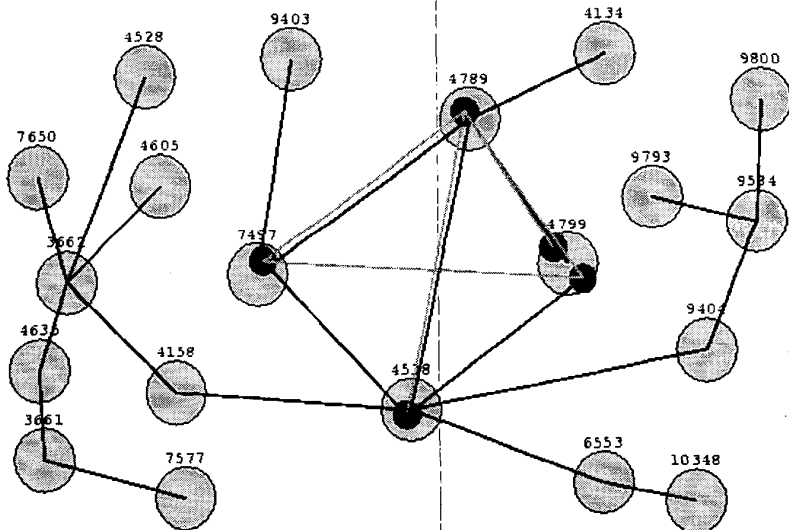


Figure 4. Topology constructed with the data collected from the backbone of the CERNET

FROM:	TO:	TIME:	TYPE:
166.111.69.75	166.111.69.91	12/24 17:21:39	BGP_UPDATE
166.111.69.91	166.111.69.75	12/24 17:21:39	BGP_NOTIFY
166.111.69.75	166.111.69.91	12/24 17:21:44	BGP_UPDATE
166.111.69.75	166.111.69.91	12/24 17:21:49	BGP_UPDATE
166.111.69.91	166.111.69.75	12/24 17:21:49	BGP_UPDATE
166.111.69.75	166.111.69.91	12/24 17:21:54	BGP_UPDATE
166.111.69.91	166.111.69.75	12/24 17:21:59	BGP_UPDATE
166.111.69.75	166.111.69.91	12/24 17:22: 4	BGP_UPDATE
166.111.69.75	166.111.69.91	12/24 17:22: 9	BGP_UPDATE

Figure 5. An improper session recorded